VoiceOps 2026 - ZentroMart voice support agent

Deepak_team - Deepak_Verma | LLM: openai/gpt-oss-120b (Groq) | Speech: Gnani STT + TTS only | Data: the synthetic ZentroMart dataset only.

1. What I built

One agent that answers from company documents, holds a conversation, and speaks. Pipeline for one turn: customer audio -> Gnani STT -> agent (LLM + retrieval + tools) -> Gnani TTS -> spoken reply. Each step reuses the previous one, and every answer logs the document chunks it used.

2. Step 1 - prompt-only baseline

A system prompt makes the LLM ZentroMart's assistant (role, limits, introduces itself as an AI assistant) and returns JSON {decision, reply}. It never sees the documents. I tried two prompts on the 52 practice questions: a *cautious* one (say 'I don't know' and offer a human when unsure) and a *guessing* one (always answer from general knowledge). Scoring uses the dataset's own rules: expected decision, must_include (';' = AND, ' / ' = OR) and must_not_include.

Run	Result on 52 practice questions
Step 1, cautious prompt	8/52 = 15.4%
Step 1, guessing prompt	not measured
Step 2, RAG	4/5 = 80.0% (PARTIAL RUN - rerun or use --s1/--s2)

The cautious prompt refuses or escalates almost everything; the guessing prompt invents policy numbers. Both fail for the same reason: the model has no access to ZentroMart's documents.

3. Step 2 - grounded answers (RAG)

- **Chunking:** each '## [SECTION-ID]' section of kb/*.md is one chunk, keeping doc type, status and effective date. Superseded and marketing text are down-ranked and labelled so precedence (policy > guide > help > FAQ > marketing) is applied; the superseded v3 returns policy is still used for electronics delivered before 1 August 2026.
- **Retrieval:** BM25 over 84 chunks, plus query expansion that bridges customer words to document words ('send it back' -> 'return'), hints from the product record (category, marketplace seller, price) and an LLM rewrite of Hindi questions into English keywords. Rules that always apply (style guide, escalation rules, business-day and date rules) are loaded into every prompt.
- **Records and tools:** products.csv rows are added when a product or SKU is mentioned. A date calculator tool does business-day (Mon-Sat, holidays excluded) and calendar-day arithmetic, but policy numbers still come from the documents, so nothing is hard-coded.
- **Honesty rule:** the prompt tells the model to say 'I don't have that information' and offer a human when the documents do not cover a question.

Answer score: 4/5 = 80.0% (PARTIAL RUN - rerun or use --s1/--s2). Gold-section recall: 62% from search alone, 100% including the always-loaded rules.

4. Step 3 - conversation

- **Memory:** the last turns, the order ID, verification state and the verified order record are kept, so 'and how long will the refund take?' is understood without repeating anything.
- **Verification (code-enforced):** order ID first (spoken groups like '8978 2837' are normalised), then last 4 phone digits or pincode. One retry, then the order is locked and no details are shared. A verification value the customer never said (including spoken digits) is rejected; the pincode and customer ID are never shown to the model.
- **Tools:** the provided lookup_order and create_ticket. A ticket is created once per conversation, only with a meaningful summary, and the agent says it exists only if created=True. Compensation demands and cancelling shipped orders are refused politely without a ticket; the agent never claims to have cancelled or approved anything.
- **Handover:** human requested, repeat contact (contacts_last_7_days), high-value damage, fraud, legal threats and safety risks create a ticket and hand over.

Test harness: replaying all 23 practice conversations (96 turns) with a scripted model passed 96/96. This validates the tools, verification, ticket creation and scoring code, not the model's judgement. A full real-model run was not completed because of the Groq free-tier daily token limit.

5. Step 4 - voice

Gnani STT (language en-IN by default, which accepts English mixed with Hindi) feeds the agent; replies are spoken with Gnani TTS (model named explicitly). The TTS language follows the reply: Devanagari -> hi-IN, Roman Hindi -> hi-en, otherwise en-IN. Replies are rewritten for the ear: amounts as 'rupees', long order numbers read in groups of four digits, markdown and section tags removed. Clips over 55 s are split. Silence or unclear audio gets a spoken 'Sorry, I didn't catch that', and a failure in the agent still produces a spoken apology, so every audio file gets a reply. One command processes a folder: python run.py --audio-dir DIR --out-dir sample_outputs.

6. What went wrong and what I fixed

- **Empty replies** on date questions (reasoning used all tokens): retry with lower reasoning effort and a bigger budget; never send silence.
- **Half-done date maths:** the model asked the calculator only for the minimum days and guessed the rest. Fixed with one range tool that returns earliest and latest date.
- **Over- and under-escalation:** a warranty question on a >10,000 INR item was escalated (my own query hint pulled in the damage rule); a cracked laptop was not. Fixed by adding the hint only for damage questions and by rules saying each escalation condition must be fully met and that escalation beats clarifying.
- **Wrong decision labels:** asking for an order ID on general questions, calling a plain 'no' a refusal, tagging helpful answers as 'clarify'. Fixed in the prompt.
- **Missing facts and script:** replies left out the seal condition or wrote 'three' for 3, and answered Hinglish in Devanagari. Fixed with explicit wording and script rules.
- **Intro:** the model sometimes skipped 'I'm the AI assistant', so the first reply is checked in code.

7. Limitations and next steps

- Retrieval is keyword-based; the expansion rules were written after reading the practice questions, so hidden-set recall will be lower. Next: add dense multilingual embeddings and compare.
- Free-tier token limits prevented re-running every evaluation after each fix; results should be re-measured on the full 52 questions and 23 conversations.
- Hindi speech is transcribed as Latin text by en-IN; I did not compare hi-IN STT or test other Hindi voices.
- Bonus work: query expansion from the product record, date calculator tool, code-level verification and ticket guards, and automatic failure-cause reports.